The impact of punctuated infectiousness on infectious disease dynamics

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Abstract

1 Introduction

The trajectory of an epidemic depends on two fundamental quantities: the basic reproduction number (R_0), which is the expected number of secondary infections that an infectious person will produce in an otherwise susceptible population; and the generation interval distribution ($g(\tau)$), which is the distribution of times at which those secondary infections are expected to occur relative to the index case's infection time ($\tau = 0$) [Wallinga and Lipsitch, 2007]. These quantities can be combined into a single object, the "infectiousness profile", defined as $A(\tau) = R_0 g(\tau)$, which describes a person's expected contribution to transmission and how it is distributed over time [Breda et al., 2012].

The reproduction number and the generation interval distribution are both expectations taken over a large population. At the individual level, the individual reproduction number (ν_i) is the expected number of infections that a particular person i will produce, such that $R_0 = E_i[\nu_i]$ [Lloyd-Smith et al., 2005]. Similarly, the "individual infectiousness profile" ($a_i(\tau)$) describes how a particular person's infectiousness is distributed over time, with $\nu_i = \int_0^\infty a(\tau)d\tau$ [Champredon and Dushoff, 2015, Svensson, 2007]. This implies the existence of an individual generation interval distribution, $\xi_i(\tau) = a_i(\tau)/\nu_i$.

Estimating R_0 and $g(\tau)$ (and thus $A(\tau)$) is often done immediately as an epidemic begins to spread. However, the shape of the individual infectiousness profile $a_i(\tau)$ is poorly understood for many pathogens. There are two main reasons for this: we lack a clear understanding of whether, and how, the shape of $a_i(\tau)$ impacts epidemic dynamics beyond what $A(\tau)$ can provide; and we lack study designs and inferential methods capable of inferring $a_i(\tau)$ independently of $A(\tau)$.

Often, a modeling choice is made to assume $a_i(\tau) \equiv A(\tau)$. However, it is likely that the full variability in $A(\tau)$ is made up of both within-individual and between-individual variation, implying $a_i(\tau)$ is generally narrower than $A(\tau)$. Emphasizing this point, some evidence suggests that the period when SARS-CoV-2 and influenza can transmit explosively is remarkably short, on the order of a few hours (**[GOYAL]**), sugesting that $a_i(\tau)$ may in some cases be much narrower than $A(\tau)$, with much of the variation in $A(\tau)$ arising from variation between individuals and on when their peak infectiousness occurs. These considerations are sometimes built into disease transmission models, but we lack a comprehensive understanding of how the punctuatedness of $a_i(\tau)$, independent of other factors, impacts disease transmission dynamics, and we lack a clear framework for incorporating $a_i(\tau)$ into the foundational disease transmission modeling frameworks upon which virtually all others are based.

At the individual level, communicating about the likely timing of a person's infectiousness is important: it is good to know if you will be very infectious for a short time, or mildly infectious for a long time, so that you can shape your behavior accordingly. Additionally, a punctuated individual infectiousness kernel could impact epidemic dynamics. Even if the epidemic has the same mean-field behavior, it could impact the stochastic dynamics via highly synchronized transmission events, especially when an outbreak is small. It could interplay with control measures, especially those designed to intervene on infectiousness at the individual level like test- or symptom-based screening and isolation. Indeed, the impact of the punctuatedness of the individual infectiousness kernel has been examined as a sensitivity analysis in various epidemiological studies. **Dan; others?** Stochastic effects are known to cascade through macroscopic features of an epidemic **Morris2025**. However, we lack a comprehensive understanding of how the sharpness of the individual infectiousness kernel, independent of other features like the individual reproduction number, basic reproduction number, and generation interval distribution, impacts epidemic dynamics under both uncontrolled and controlled settings. Furthermore, we lack an understanding of how to measure the shape of the individual infectiousness kernel reliably, along with the settings when the kernel's shape

is statistically identifiable.

The main goal: holding $A(\tau)$ constant, how do different choices of $a_i(\tau)$ impact epidemic dynamics? And how would we know what $a(\tau)$ looks like?

2 The individual infectiousness profile

2.1 Expressing the Hudson–Kermack–McKendrick model in terms of individual infectiousness

Early work by Hudson, and later by Kermack and McKendrick [Kermack and McKendrick, 1927], modeled the transmission of a permanently immunizing infection using an integral equation. In modern notation [Breda et al., 2012], the force of infection is

$$F(t) = \int_0^\infty F(t-\tau)S(t-\tau)A(\tau)d\tau \tag{1}$$

where $F(t)$ is the force of infection at time t, $S(t)$ is the density of susceptible individuals in the population at time t, and $A(\tau)$ is the infectiousness profile describing the expected contribution to the force of infection from an individual who was infected τ time units ago. The incidence of disease is the product of the force of infection and the susceptible density (*i.e.*, $-\frac{dS}{dt} = F(t)S(t)$); thus, the infectiousness profile $A(\tau)$ is the fundamental object that determines how an epidemic unfolds. The infectiousness profile is related to the basic reproduction number R_0 and the generation interval distribution $g(\tau)$:

$$R_0 = \int_0^\infty A(\tau)d\tau \tag{2}$$

$$g(\tau) = \frac{A(\tau)}{R_0} \tag{3}$$

Since $A(\tau)$ is an expectation, it does not necessarily reflect any single person's infectiousness, just as ν_i can differ substantially from R_0. For example: the SEIR model is conceptually motivated by considering an individual who undergoes a latent period of length $x_i \sim \text{Exp}(\gamma)$ during which they are not infectious, followed by an infectious period of length $y_i \sim \text{Exp}(\alpha)$ during which they are infectious at level β. Mathematically, this "stepwise" individual infectiousness profile is expressed as

$$a_i(\tau) = \begin{cases} \beta & \tau \in [x_i, x_i + y_i] \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

This yields the population-level infectiousness kernel

$$A(\tau) = E_i[a_i(\tau)] = \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} a_i(\tau) = \beta\frac{\gamma}{\gamma-\alpha}(e^{-\alpha\tau} - e^{-\gamma\tau}) \tag{5}$$

i.e., a smooth, bell-shaped curve that arises from a sum of discontinuous, stepwise $a_i(\tau)$ (**Figure XX**). Furthermore, other choices of $a_i(\tau)$ can yield the same $A(\tau)$, including

$$a_i(\tau) = \beta\frac{\gamma}{\gamma-\alpha}(e^{-\alpha\tau} - e^{-\gamma\tau}) \tag{6}$$

a "smooth" individual infectiousness profile where each person's infectiousness is identical to the population-level average, and

$$a_i(\tau) = \frac{\beta}{\gamma}\delta_{t_i}(\tau) \tag{7}$$

a "spike" infectiousness profile where $\delta_i(\tau)$ is a delta function with mass at time t_i, and t_i is distributed according to the generation interval density $A(\tau)/R_0$ (**Figure XX**). In this case, a person's infectiousness is concentrated at a single moment. In general, any definition of $a_i(\tau)$ such that $E_i[a_i(\tau)] = A(\tau)$ will yield the same mean-field population-level epidemic dynamics; however, they may have important stochastic consequences (**Figure XX**), which are the main topic of this article.

2.2 Properties and components of the individual infectiousness profile

Like $A(\tau)$, $a_i(\tau)$ is related to the individual reproduction number ν_i and the individual intrinsic generation interval distribution $\xi_i(\tau)$:

$$\nu_i = \int_0^\infty a_i(\tau) d\tau \tag{8}$$

$$\xi_i(\tau) = a_i(\tau)/\nu_i \tag{9}$$

Re-arranging Eqs. 8 - 9,

$$a_i(\tau) = \nu_i \xi_i(\tau) \tag{10}$$

So far, this assumes that interpersonal contacts are uniform over time and across the population – or else that any non-uniformity is captured in ν_i and/or $\xi_i(\tau)$. A futher decomposition is possible by introducing an explicit contact process $c_i(t)$, defined with respect to the uniform reference contact process $\bar{c}$ against which both R_0 and ν_i are defined. Let $c_i(t)$ be a nonnegative function describing person i's proportional deviations from $\bar{c}$ (so $c_i(t) \equiv 1$ represents the reference process), and let $b_i(\tau)$ be a density capturing the timing of person i's secondary infection attempts under $\bar{c}$. Then,

$$a_i(t_i, \tau) = \nu_i b_i(\tau) c_i(t_i + \tau) \tag{11}$$

where t_i is person i's infection time and τ represents time since infection. For ν_i to retain its meaning, we require $E_t[c_i(t)] = 1$, that is, for a randomly chosen time t, the expected value of person i's contact modulator is 1. This is satisfied, for example, by a periodic function with average value 1 or by a stationary random process with mean 1. Another way of interpreting the contact modulator is that $\nu_i c_i(t)$ is the expected number of people that person i would infect if all their infectiousness were concentrated at time t (*i.e.*, if b_i were a delta function with mass at time t). The density $b_i(\tau)$ therefore reflects the "biological" component of a person's infectiousness timing, and $c_i(t)$ captures the "contact-related" component of a person's infectiousness timing, while ν_i modulates the person's total infectiousness.

This extends back to the population-level model. If we assume that c_i is independent of ν_i and b_i, but allow ν_i and b_i to depend on one another (for example, we might expect a person with high overall infectiousness to also have a longer duration of infectiousness), then the population-level infectiousness profile is

$$A(t_i, \tau) = E_i[a_i(\tau)] = E_i[\nu_i b_i(\tau)] E_i[c_i(t_i + \tau)] \tag{12}$$

with the expectation taken over the population indexed by i. We define $C(t) = E_i[c_i(t)]$ to be the population-mean contact modulator at calendar time t. We define $B(\tau) = E_i\left[b_i(\tau)\frac{\nu_i}{R_0}\right]$, *i.e.*, the population-average biological infection timing density weighted by the person's infectiousness relative to the population mean. With these definitions, we obtain

$$A(t_i, \tau) = R_0 B(\tau) C(t_i + \tau) \tag{13}$$

Note that, like $c_i(t)$, $C(t)$ has time-average 1, since $E_t[C(t)] = E_t[E_i[c_i(t)]] = E_i[E_t[c_i(t)]] = 1$ (Fubini). If $C(t) \equiv 1$, we recover the original expression of $A(\tau) = R_0 B(\tau) = R_0 g(\tau)$, where $B(\tau)$ becomes the intrinsic generation interval distribution.

3 The impact of the individual infectiousness profile on uncontrolled epidemics

3.1 Punctuated individual infectiousness profiles generate later and more variably-timed epidemics.

Even in large, well-mixed populations, stochasticity can durably impact the trajectory of an epidemic. One way this stochasticity enters is via a random time shift, where epidemics take a variable amount of time to reach the exponential growth phase [Morris et al., 2024], leading to variation in the timing of the entire epidemic trajectory. Here, we address the question of how this time shift distribution depends on the shape of the individual infectiousness profile, holding all else equal.

Simulations confirm that narrower infectiousness profiles ($\psi \to 0$) yield more variable and later-shifted epidemic timings (**Figure XX**, **Supplementary Table Sxx**). Across 10,000 simulated epidemics in a population of size 1000 using SARS-CoV-2 omicron-like parameters, the median time to reach 100 infections was XX days for $\psi = 0$ (IQR: [XX, XX]) vs. XX days for $\psi = 1$ (IQR: [XX, XX]). The impact of ψ on epidemic timing was lessened with lower R_0 (influenza-like parameters, median time to 100 infections: XX days for $\psi = 0$ [IQR: XX, XX] vs. XX days for $\psi = 1$ [IQR: XX, XX]) and increased with higher R_0 (measles-like parameters, median time to 100 infections: XX days for $\psi = 0$ [IQR: XX, XX] vs. XX days for $\psi = 1$ [IQR: XX, XX]). From another perspective: by day XX of the simulated epidemic under measles-like parameters, XX% of the outbreaks with $\psi = 1$ had reached 100 infections, while only XX% of the outbreaks with $\psi = 0$ had reached 100 infections (**Figure XX**).

These observations are confirmed by the underlying theory. The cumulative incidence over time, $Z(t)$, during the early exponential growth phase of an epidemic can be approximated as a deterministic exponential growth function with a random initial condition scaling factor W:

$$Z(t) \approx W \cdot ce^{rt} \tag{14}$$

with exponential growh rate r, and where c is the mean-field initial condition (when $W = 1$).

With a change of variables, the cumulative incidence can be expressed as the same deterministic growht function with a random time shift ς:

$$Z(t) \approx ce^{r(t+\varsigma)} \tag{15}$$

where

$$\varsigma = \frac{\log W - \log E[W]}{r} \tag{16}$$

Here, $E[W]$ is the initial condition and r is the exponential growth rate, so $\varsigma \propto \log[W]$.

First, we consider the variance of the time shift, $Var[\varsigma]$. For ease of notation, we focus on $Var[W]$, noting that $Var[\varsigma]$ is monotonically related to $Var[W]$. It can be shown that **Supplementary Text**

$$Var[W] = m_2 + R_0^2 \rho^{2\psi\alpha} [\rho_2^{(1-\psi)\alpha} - \rho^{2(1-\psi)\alpha}]$$

which is maximized when $\psi = 0$ (the infectiousness profile is narrow) and becomes 0 when $\psi = 1$ (the infectiousness profile is maximally broad). (Can we say something about how this suggests that the

variance of the time shift scales roughly linearly with ψ?)

Next, we consider the typical time shift, which we obtain by studying the mode of the distribution of W.

We can approximate the distribution of W using moment matching to a generalized Gamma distribution. With the Gamma-shaped individual infectiousness profiles, the moments have closed-form expressions. Using two-moment matching to a Gamma distribution, we find that

$$\text{mode}_\varsigma = \frac{1}{r}\log(1 - \text{CV}^2_{\text{cond}})]$$

Here, (description). Again, we see that the mode shifts later as $\psi \to 0$.

The theory aligns well with the simulations. Using generalized gamma distributions fit using moment matching to the first five moments of W, we get similar survival curves for the time to 100 infected (**Figure XX**).

3.2 Narrower individual infectiousness profiles yield greater uncertainty in epidemic growth rate estimates.

Based on simulations.

3.3 Narrower biological infectiousness profiles interact with periodic contact patterns to generate overdispersion in secondary infections.

Closed-form solution:

$$k = \frac{2}{\bar{c}^2}A^2\left(1 + \frac{\omega^2}{r^2}\right)^{\psi\alpha} = \frac{2}{\bar{c}^2}A^2\left(1 + (\frac{z^*_{per}}{z_{per}})^2\right)^{\psi\alpha}$$

Figure 1: Impact of the individual infectiousness profile on epidemic timing. Survival curves for time-to-100. growth rate estimate box-and-whiskers. Overdispersion/k vs psi heatmap.

4 The impact of the individual infectiousness profile on detect-and-isolate interventions

- The biologically anchored detection model
- For fixed detection offset, TE depends on psi. Skewed profiles buy a bit of detection time after the peak where TE is still greater for spikier profiles (within reason).
- Symptoms arising somewhat randomly smooth out the effect. For spikier profiles, they boost TE when mean detection is after the peak infectiousness and reduce TE when mean detection is beffore peak infectiousness – meaningfully.
- Regular screening: TE is sensitive to screening frequency and delays in response, especially in specific regions of sensitivitiy-frequency-window size combinations.
- TE is only part of the story: isolation shrinks generation intervals (moreso for smooth profiles), and increases overdispersion (moreso for spike profiles), which has an impact on epidemic dynamics that differ from what you'd expect if you simply ran an epidemic with $R_{\text{eff}} = R_0(1 - TE)$. Give some simulations to demonstrate.

5 The impact of individual infectiousness on gathering size restrictions

Briefly, we also consider epidemic control by restricting gathering sizes.

- The theory
- The model
- Like with detect-and-isolate, the reduction in R_{eff} isn't the whole story: gathering size restrictions impact overdispersion, and again do so more for spike than for smooth profiles.

6 Notes on detect and isolate

We first consider the detect-and-isolate interventions. Detection may occur through various mechanisms, e.g., the onset of symptoms or a positive diagnostic test.

We can express the effectiveness of a detect-and-isolate intervention in terms of the *test effectiveness* [Middleton and Larremore, 2024].

If the detection mechanism is independent of the individual infectiousness profile (e.g., detection is anchored to the infection time but does not depend on what happens thereafter), the imapct of the individual infectiousness profile washes out (**Supplementary Text**). However, if detectability is anchored somehow to the infectiousness profile – which is likely to be the case in general – the individual profile begins to matter.

Here, we consider detection strategies that are anchored to the mode of the individual infectiousness profile (i.e., the time of peak infectiousness). Let $D(\tau)$ be the cumulative probability that a person has been detected by time τ.

To begin, we consider a fixed, deterministic detection time relative to the mean - this might correspond to symptoms arriving deterministically some amount of time prior to (or after) peak infectiousness.

Maybe: just show a TE profile for a spike, smooth, and $\psi = 0.5$ profile for a deterministic offset (size of offset on the horizontal axis, TE on the vertical); and then show the analogous curves for when the detection profile is random (say, normal, where now the horizontal axis captures the mean of the distribution, and we pick some reasonable standard deviation). This should illustrate how the timing of detection (with variation) interplays with the punctuation of the infectiousness profile. That might be enough here.

Then we can focus on the implications for the generation interval distribution and secondary infection distribution.

I want to try my hand at making this figure: what do we need?

- The general detection distribution framework
- Narrower infectiousness profiles transfer detection into an all-or-nothing proposition
- Detection and isolation shorten the generation interval distribution, moreso for smoooth profiles, which means that epidemics may grow (and shrink) faster than a naive application of TE would suggest...
- while detection and isolation increases overdispersion in secondary infections, moreso for spike profiles, so that epidemics are less likely to take off (but more explosive) than would be expected with a naive application of TE.
- The takehome: detection and isolation reduce R_{eff} via TE, but a simple application of TE won't tell us everything we need to know: instead, we need to know something about the individual infectiousness profile to properly anticipate how detect-and-isolate interventions will impact epidemic dynamics.
- Furthermore: narrower infectiousness profiles make detect-and-isolate interventions sensitive to small shifts, especially near the infectiousnesss profile itself. So, if symptoms move from slightly after to slighlty before infectiousness on average, this will have an outsize impact on narrow profiles vs. smooth ones. Similarly, if a small increase in testing cadence could have a huge impact for spike profiles, if we shift the mean of D from slightly after to slightly before peak infectiousness; but in other areas, it'll hardly have any impact on spike profiles, whereas the change in effectiveness for smooth profiles should be more gradual.
- This last point is relevant for when detection is anchored to the mode (or otherwise anchored to the individual infectiousness profile); otherwise, using the population average I think should be fine (e.g. if detection depends only on time of infection but not subsequent infectiousness).
- The takeaway from this: if we want to know the impact of shifting biology (changes in the symptom timing) or shifts in recommendations (changes in the cadence of testing), we should know something about the individual infectiousness profile (and how it relates to detectability).
- Adding in delays/turnaround time will also impact all this.

7 Identifiability and inference of the individual infectiousness profile's shape

While the concentration of the individual infectiousness profile can impact how an epidemic unfolds, we lack frameworks for estimating it. Many approaches exist for estimating the generation interval, but there has not been a systematic approach developed to assess how individual infectiousness varies and to specify where on the continuum we sit—i.e., how much of the variation in the generation interval distribution is attributable to variation between individuals, and how much to variation in infection timing within individuals.

We begin by assessing under which conditions we can hope to infer the shape of the individual infectiousness profile. We restrict to gamma-distributed timings, and consider cases where we assume that κ is shared across individuals and where κ comes from a beta distribution governed by hyperparameters α and β. Then, when κ is identifiable, we introduce a hierarchical Bayesian inference strategy to infer the shape of the individual infectiousness profile given contact tracing data.

For this section: let's imagine that there's a single *psi* value shaed among everyone. We can use a Beta prior. Two things to consider: how do we estimate ψ given a list of infection times (assume we know them exactly)? And how do we estimate ψ given infection times and knowledge of $c_i(t)$? My hope is that this will emphasize that knowing the "negatives" – i.e., the contacts that "should" have led to transmission but did not – improves our inferential power substantially. Then, I want to think about how to achieve a given power in each scenario across R0, assuming some fixed generation interval distribution. That should be enough for now. We should probably include some amount of uncertainty in infection times for this base analysis, and we should post code for running the analysis given a line list of infection times and contacts. In the Discussion, we need to address the fact that there could be different ψ_i values across individuals, themselves drawn from a distribution, so we'd want some kind of hierarhcical Bayesian model for inference here; also, we don't consider differences in the height (i.e. inherent biological differences in ν_i / b_i), which is another key thing to include in the inference.

8 Discussion

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Supplementary Information

Supplementary Methods

Supplementary Figures

Supplementary Tables